



Build a Virtual Private Cloud (VPC)



Farhan Rasheed

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me



Introducing Today's Project!

In this project, I will demonstrate how to design and configure an AWS Virtual Private Cloud (VPC), create subnets, and deploy an Internet Gateway to enable secure internet connectivity for resources within the VPC.

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is a logically isolated virtual network in AWS where you can launch and manage AWS resources such as EC2 instances in a secure and controlled environment. It allows you to define your own IP address range, create subnets, configure route tables, and control inbound and outbound traffic using security groups and network ACLs. Amazon VPC gives you full control over your cloud networking, similar to a traditional on-premises data center. It improves security by isolating resources, allows private and public networking, enables secure connectivity to on-premises networks using VPN or Direct Connect, and helps design scalable, flexible, and highly available cloud architectures.

In today's project, I used Amazon VPC to create a private network of my own. Created subnets then attached an internet gateway to my vpc so that i can launch resources.

Personal reflection

This project took me around 30 mins.



I was surprised by how simple and enlightening this project turned out to be.



Virtual Private Clouds (VPCs)

What I did in this step

In this step, I will configure a Virtual Private Cloud (VPC) to provide a dedicated and isolated networking environment for the cloud infrastructure deployed in this project.

How VPCs work

VPCs are virtual private cloud. They are the reason why resources can be made private and also have a control on them like how they should be designed and how they will be connected to the internet and such.

Why there is a default VPC in AWS accounts

This default VPC is there so that we could launch resources (e.g. EC2 instances) and connect services together from Day 1 of using AWS.



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NextWork VPC

IPv4 CIDR block [Info](#)
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☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Defining IPv4 CIDR blocks

CIDR (which stands for Classless Inter-Domain Routing) is a way to assign a whole block of IP addresses, kind of like creating a zone/area in a city.



Subnets

What I did in this step

In this step, I will create a subnet inside the VPC to logically segment the network and provide a dedicated IP address range for AWS resources.

Creating and configuring subnets

subnets are like different neighborhoods inside your city. We use subnets to group resources with similar access rules and restrictions. Some subnets might be public areas that all resources can access (public subnets) while others are private areas

Public vs private subnets

The subnet is not considered a public subnet because it does not have a route to an Internet Gateway. A subnet becomes public only after it is associated with a route table that directs internet-bound traffic (0.0.0.0/0) to an Internet Gateway attached to the VPC.



Enable auto-assign public IPv4 address

Subnets (1/1) [Info](#)

Last updated less than a minute ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

Subnet ID: subnet-06f5abd29f009677a [Clear filters](#)

☒	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☒	Public 1	subnet-06f5abd29f009677a	Available	vpc-07213453076ae8259 Nex...	Off	10.0.0.0/24

Auto-assigning public IPv4 addresses

When you enable auto-assign public IPv4 address for a subnet, any EC2 instance launched in that subnet will instantly get a public IP address so we won't have to create one manually



Internet gateways

What I did in this step

In this step, I will deploy and attach an Internet Gateway (IGW) to the VPC. The Internet Gateway serves as a bridge between the VPC and the internet, allowing resources in public subnets to send and receive internet traffic.

Setting up internet gateways

Internet Gateways (IGWs) are AWS networking components that allow communication between a Virtual Private Cloud (VPC) and the internet. Think of an Internet Gateway as a doorway that connects your private AWS network (VPC) to the public internet.

Attaching an Internet Gateway (IGW) to a VPC enables communication between resources in the VPC and the internet.



Internet gateway igw-06c6676458b259375 successfully attached to vpc-07213453076ae8259

igw-06c6676458b259375 / NextWork IG

Actions

Details

Info

Internet gateway ID	State	VPC ID	Owner
igw-06c6676458b259375	Attached	vpc-07213453076ae8259 NextWork VPC	582577265973



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